

Key concepts for STAT 154 / 254

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Key concepts

This class will focus almost entirely on *supervised learning*.

Here are the key concepts:

- ▶ What does it mean to do well on a supervised learning problem?
 - ▶ Random data, loss, and risk
 - ▶ **Empirical risk minimization, generalization**
- ▶ What would we do if we had an infinite amount of data?
 - ▶ Generating computational tools to fit arbitrary functions
 - ▶ **Expressive function classes**
- ▶ What goes wrong when we have only a limited amount of data?
 - ▶ Avoid overfitting by limiting the expressivity of function classes
 - ▶ **Regularization**
- ▶ How do we find a sweet spot between expressiveness and overfitting?
 - ▶ Theory: **generalization bounds**
 - ▶ Practice: **out-of-sample error**

QSAR Fish Toxicity Dataset

These examples will be based on the UCI QSAR Fish Toxicity Dataset.

The dataset consists of a set of 908 chemicals (one row each), with data for:

- ▶ LC50: the concentration that causes death in 50% of test fish over a test duration of 96 hours.
- ▶ MLOGP: (molecular properties)
- ▶ CIC0: (information indices)
- ▶ GATS1i: (2D autocorrelations)
- ▶ NdsC: (atom-type counts)
- ▶ NdsCH: (atom-type counts)
- ▶ SM1_Dz: (2D matrix-based descriptors)



Figure 1: Sad fish

“Quantitative structure–activity relationship” (QSAR) models predict some chemical property (toxicity to the fathead minnow) from structural features of the molecules.

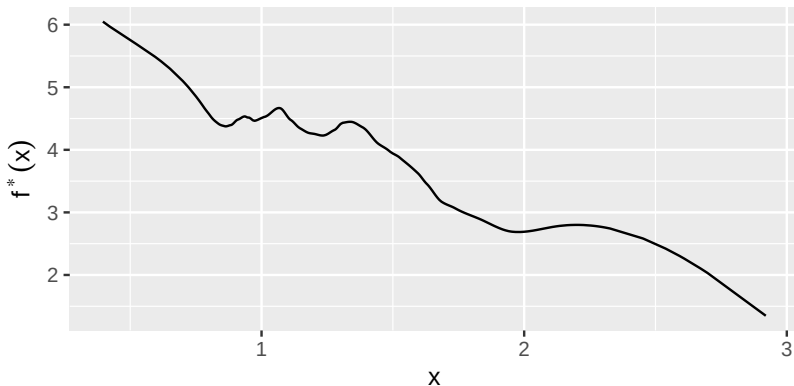
Task: Set $x = \text{GATS1i}$, $y = \text{LC50}$, and use ML to predict fish mortality from 2D autocorrelations.

i Question

Describe some concrete settings when you would want to use an ML model that could perform this task.

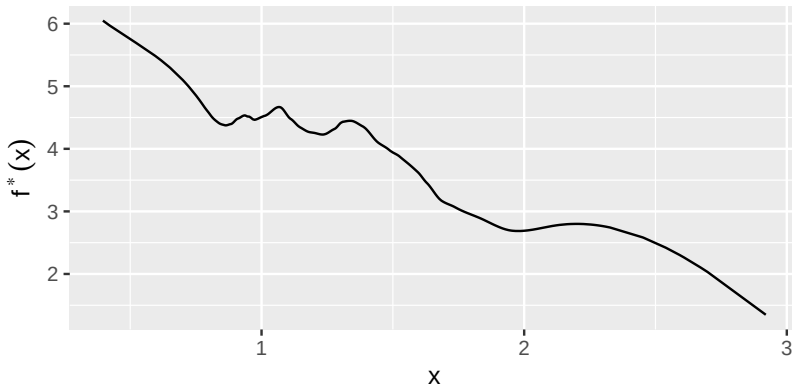
Note: I will be simulating data using a “ground truth” that is loosely based on patterns from the actual data.

Fitting a function



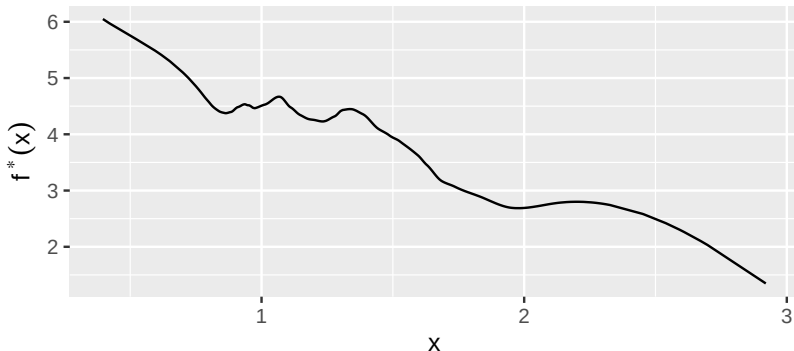
Let's imagine that y (fish toxicity) is determined exactly by x (2D autocorrelations).
That means $y = f^*(x)$ for some f^* . **(I will drop this assumption later in the lecture.)**

Fitting a function



Let's imagine that y (fish toxicity) is determined exactly by x (2D autocorrelations). That means $y = f^*(x)$ for some f^* . **(I will drop this assumption later in the lecture.)** We don't know f^* , but we can *query* it by selecting x (a molecule) and measuring y (seeing what concentration kills minnows)

Fitting a function



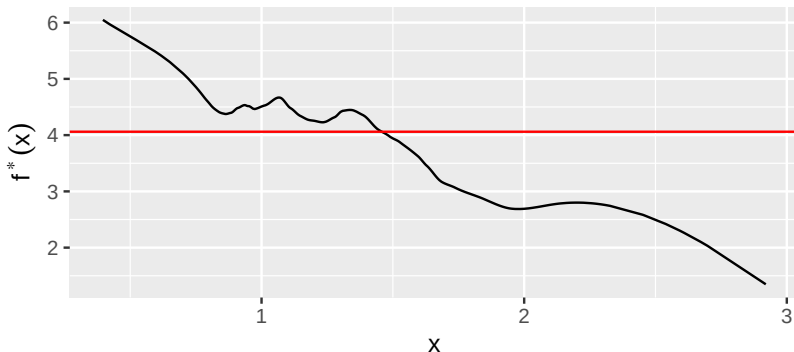
Task: Query (x_n, y_n) pairs and make a *program* that returns $\hat{f}(x) \approx f^*(x)$.

Question

Suppose you can query a very large number of datapoints.

- ▶ Can you ever perfectly approximate $f^*(x_{\text{new}})$ at an unseen x_{new} ?
- ▶ Propose the simplest procedure you can think of (even if not very good).
- ▶ Propose a more complex procedure that will do a better job.

Constant functions

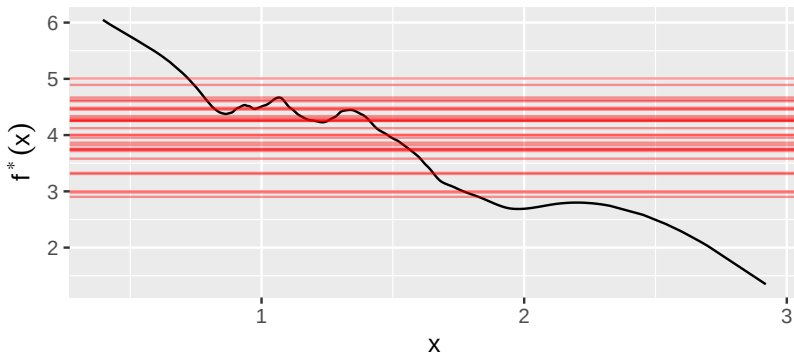


The simplest approximation is arguably the constant $\hat{f}(x) = \frac{1}{N} \sum_{n=1}^N f^*(x_n)$.

Question

- ▶ Think of an application for which a good constant is enough.
- ▶ Think of an application for which a good constant is not enough.
- ▶ Will the constant you estimate depend on which x_n you select? How?
- ▶ Can you think of other ways to take the data and choose the constant?

Constant functions



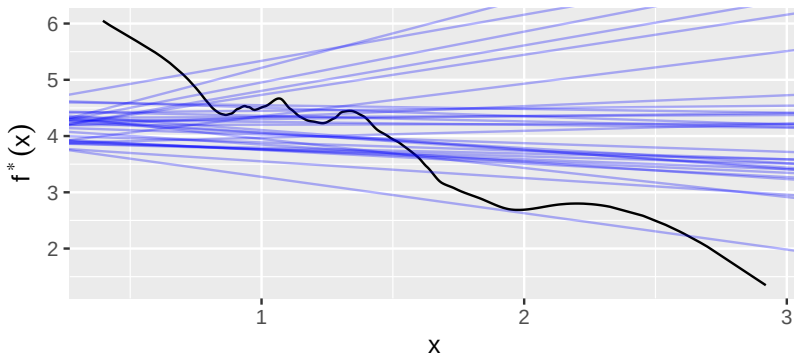
Using a constant function selects $\hat{f}$ from the *function class*:

$$\mathcal{F}_{\text{constant}} := \{f : f(x) = c \text{ for some real-valued } c\}$$

Our “estimation procedure” takes in data $\mathcal{D} = \{(x_1, y_1), \dots, (x_N, y_N)\}$ and returns a function $\hat{f} \in \mathcal{F}_{\text{constant}}$. This is equivalent to choosing a $\hat{c}$ and taking $\hat{f}(x) = \hat{c}$.

For a procedure in general, $\mathcal{F}$ is the set of *all functions that it could represent*.

Linear functions



We can also define the class of linear functions:

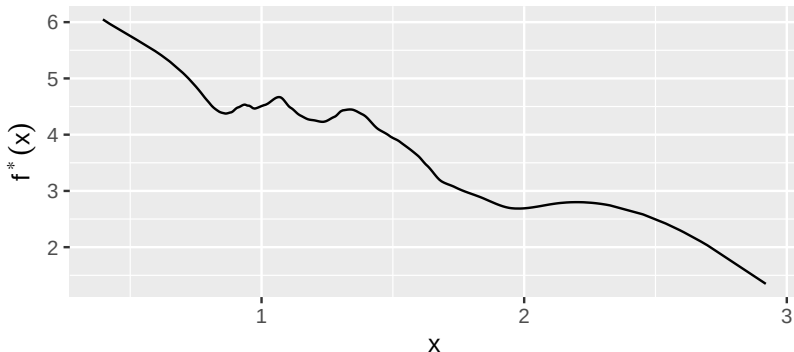
$$\mathcal{F}_{constant} := \{f : f(x) = c \text{ for some real-valued } c\}$$

$$\mathcal{F}_{linear} := \{f : f(x) = c + bx \text{ for some real-valued } c, b\}$$

Every constant function is also a linear function, but some linear functions are not constant functions.

We say that $\mathcal{F}_{linear}$ is a *more expressive function class* than $\mathcal{F}_{constant}$.

Function classes

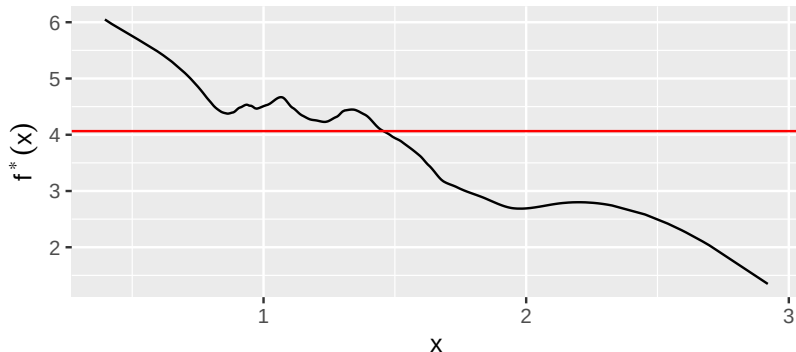


A key theme of this class is developing expressive function classes!

i Question

Identify the function classes for some of the examples given earlier. Which are more expressive? Which are the least expressive?

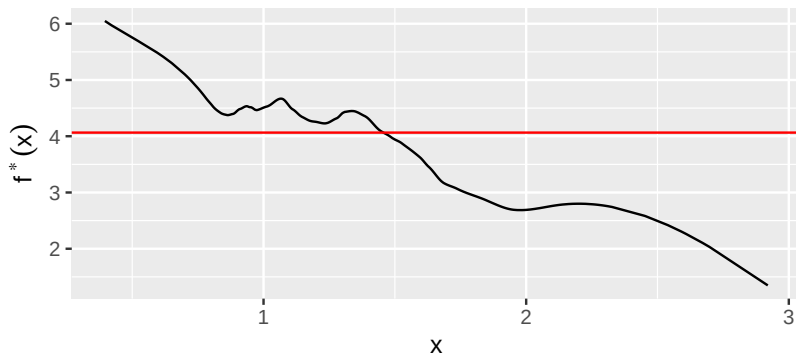
Loss



We want to choose $\hat{f} \in \mathcal{F}$ that is a “good approximation.”

Suppose we have some x with associated y . Let the “loss function” $\mathcal{L}(f(x), y)$ measure how “bad it is” to predict $f(x)$ when the truth was y .

Loss



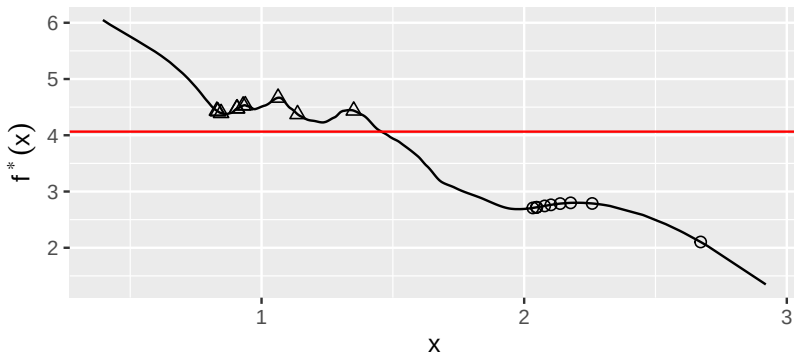
Common examples:

- ▶ $\mathcal{L}(f(x), y) = (f(x) - y)^2$
- ▶ $\mathcal{L}(f(x), y) = |f(x) - y|$
- ▶ $\mathcal{L}(f(x), y) = I(f(x) \neq y)$ (Here, $I(\cdot)$ is the “indicator function.” It evaluates to 1 if its argument is true, and 0 otherwise.)

i Question

What are some properties any loss function should satisfy?

Risk

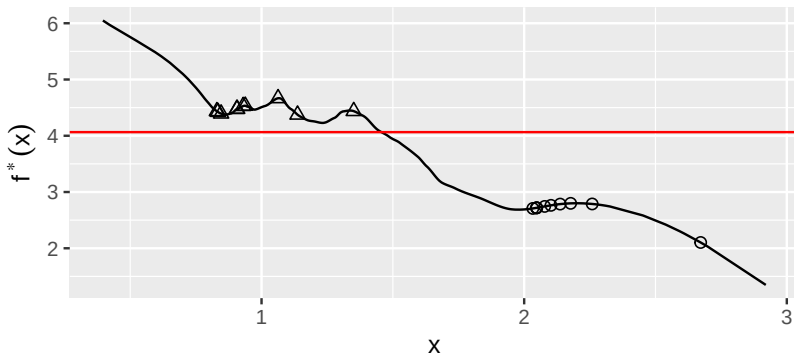


The loss is defined at a single datapoint. Unless we really care about only one prediction, it makes sense to *average* the loss over multiple datapoints:

$$\frac{1}{N} \sum_{n=1}^N \mathcal{L}(f(x_n), y_n)$$

Note that, for a given f , the average loss will be different depending where the points are!

Risk

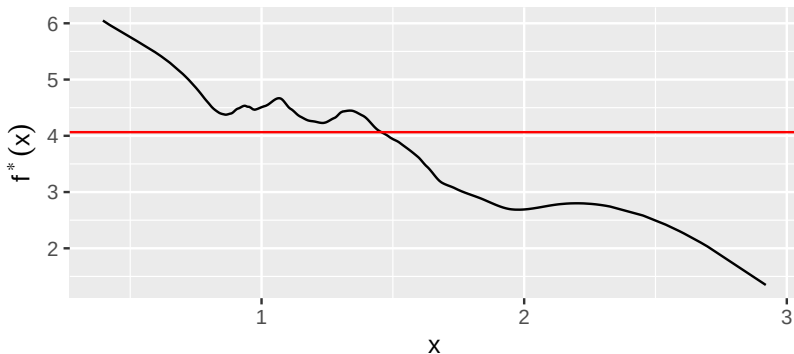


If we believe that future x will be drawn randomly, independently, and identically distributed (IID) from a future distribution $p(x)$, then it makes sense to measure the *average loss*.

This average loss is called “risk”.

$$\mathcal{R}(f) := \mathbb{E}_{p(x,y)} [\mathcal{L}(f(x), y)] \quad \text{”Risk”}$$

Note that risk depends only on the function f — the distribution or data is considered fixed. **Different distributions give different risks.**

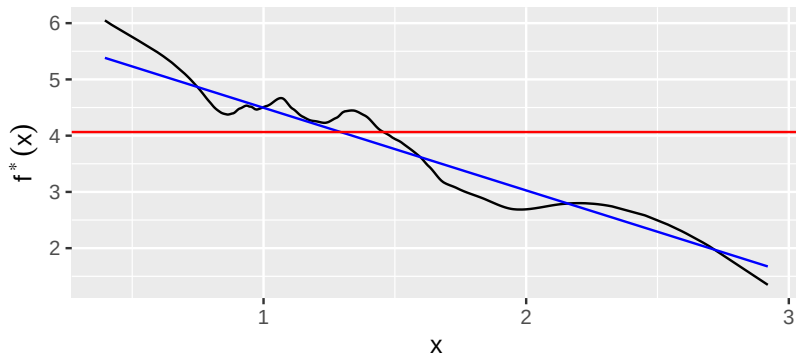


i Question

Recall our QSAR application.

- ▶ Describe a use case where the IID assumption might be reasonable
- ▶ Describe a use case where the IID assumption might not be reasonable
- ▶ What does the IID assumption imply about the relationship between the other chemical variables?
- ▶ Does minimizing $\mathcal{R}(f)$ guarantee an accurate prediction on any particular x ?

Risk



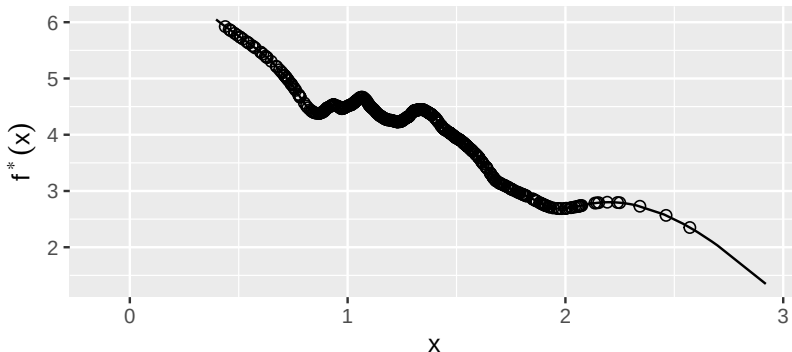
There's usually a gap between the best function f^* and the best function in our class:

$$\bar{f} := \operatorname{argmin}_{f \in \mathcal{F}} \mathcal{R}(f) \quad f^* := \operatorname{argmin}_f \mathcal{R}(f) \quad \mathcal{R}(\bar{f}) - \mathcal{R}(f^*) \geq 0$$

This gap is typically *lower* for *more expressive function classes*.

Why not just make our function classes as expressive as possible?

Risk



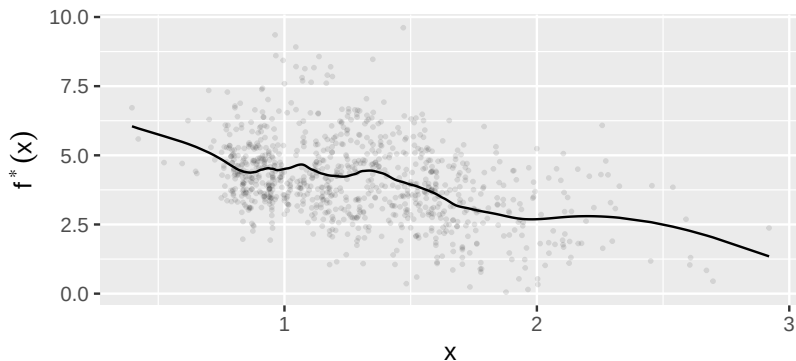
If we want to learn a function that makes $\mathcal{R}(f)$ small, we should draw training data $x_n \stackrel{\text{iid}}{\sim} p(x)$ and form the “empirical risk:”

$$\mathcal{R}(f) := \mathbb{E}_{p(x,y)} [\mathcal{L}(f(x), y)] \quad \text{”Risk”}$$

$$\hat{\mathcal{R}}(f) := \frac{1}{N} \sum_{n=1}^N \mathcal{L}(f(x_n), y_n) \quad \text{”Empirical risk”}.$$

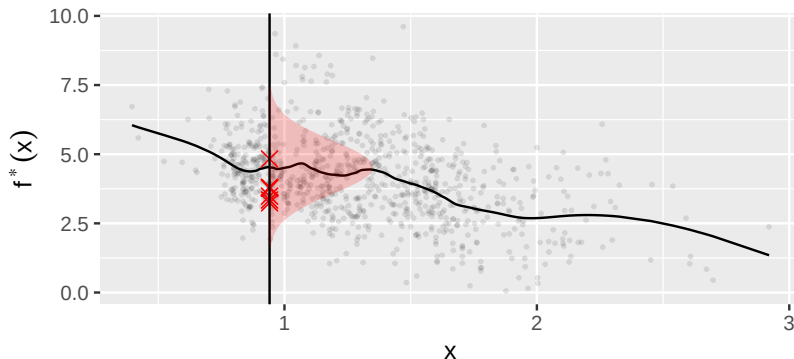
Trying to make $\mathcal{R}(f)$ small by minimizing $\hat{\mathcal{R}}(f)$ is called **empirical risk minimization** (ERM). ERM is the main idea in this course.

Response noise



Up to now, we've been imagining that $y = f^*(x)$. But usually our data looks like this.

Response noise

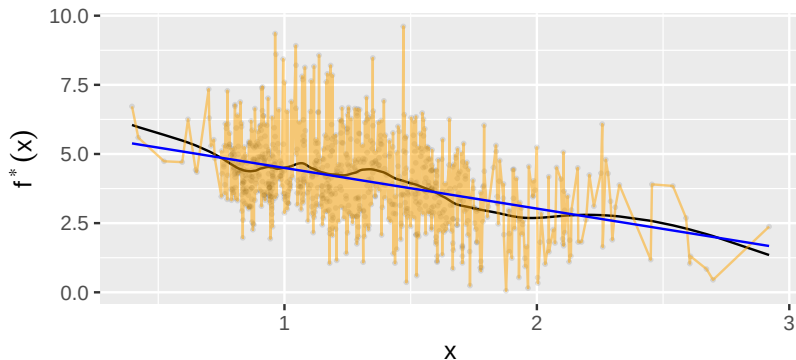


If you query the same x and get different values, you know that there is *randomness in y* . Even if you don't observe repeats, if “nearby” x give very different y , you might think of y as random.

If y is random, then at each x there is a conditional distribution, $p(y|x)$, which is different for each x .

Note that risk is the expectation over both x and y : $\mathcal{R}(f) := \mathbb{E}_{p(x,y)} [\mathcal{L}(f(x), y)]$.

Response noise



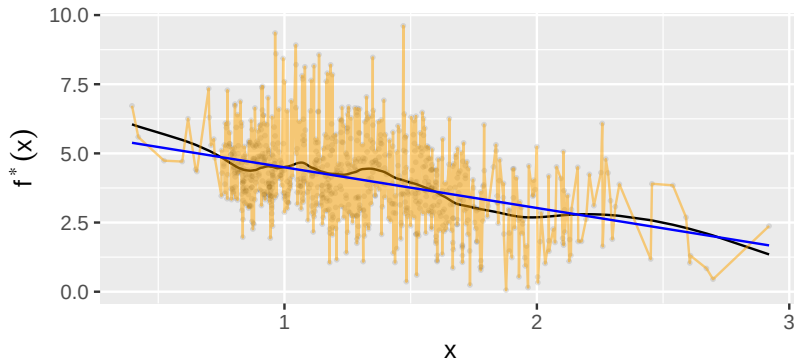
ERM finds the best function in our function class to the *actual data*:

$$\hat{f} := \operatorname{argmin}_{f \in \mathcal{F}} \hat{\mathcal{R}}(f)$$

The orange line is your $\hat{f}$ if $\mathcal{F}$ is *very expressive*!

If all you see is the data points, how can you tell which is the better prediction function?

Response noise

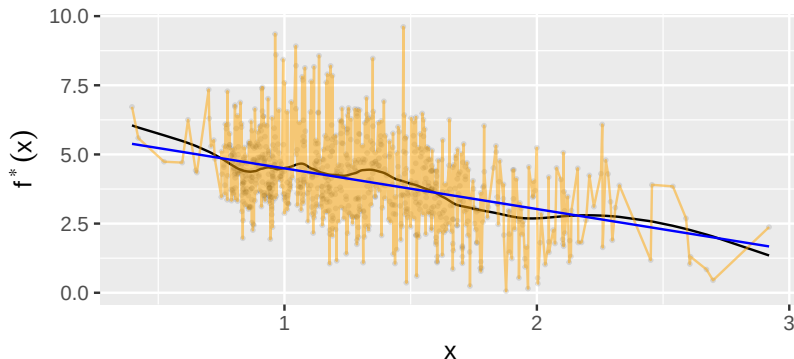


If the function class $\mathcal{F}$ is too expressive, we may fit the noise, not the risk.

$$\hat{f} := \operatorname{argmin}_{f \in \mathcal{F}} \hat{\mathcal{R}}(f) \quad \bar{f} := \operatorname{argmin}_{f \in \mathcal{F}} \mathcal{R}(f) \quad \Rightarrow \quad \mathcal{R}(\hat{f}) - \mathcal{R}(\bar{f}) \geq 0$$

This gap tends to be *higher* for *more expressive function classes*.

A fundamental tradeoff



$$\hat{f} := \operatorname{argmin}_{f \in \mathcal{F}} \hat{\mathcal{R}}(f) \quad \bar{f} := \operatorname{argmin}_{f \in \mathcal{F}} \mathcal{R}(f) \quad f^* := \operatorname{argmin}_f \mathcal{R}(f)$$

“Regret” is the gap between what we learn and what we might have learned. Its two constituent terms go in opposite directions with the expressivity of the function class:

$$\text{Regret} := \mathcal{R}(\hat{f}) - \mathcal{R}(f^*) = \underbrace{\mathcal{R}(\hat{f}) - \mathcal{R}(\bar{f})}_{\text{Expressivity } \uparrow} + \underbrace{\mathcal{R}(\bar{f}) - \mathcal{R}(f^*)}_{\text{Expressivity } \downarrow}$$

$$\hat{f} := \operatorname{argmin}_{f \in \mathcal{F}} \hat{\mathcal{R}}(f) \quad \bar{f} := \operatorname{argmin}_{f \in \mathcal{F}} \mathcal{R}(f) \quad f^* := \operatorname{argmin}_f \mathcal{R}(f)$$

$$\text{Regret} := \mathcal{R}(\hat{f}) - \mathcal{R}(f^*) = \underbrace{\mathcal{R}(\hat{f}) - \mathcal{R}(\bar{f})}_{\text{Expressivity } \uparrow} + \underbrace{\mathcal{R}(\bar{f}) - \mathcal{R}(f^*)}_{\text{Expressivity } \downarrow}$$

In this class we will,

- ▶ Learn to make and fit expressive $\mathcal{F}$ so that $\mathcal{R}(\bar{f}) - \mathcal{R}(f^*)$ is small,
- ▶ Learn how to make an $\mathcal{F}$ less expressive so that $\mathcal{R}(\hat{f}) - \mathcal{R}(f^*)$ is small,
- ▶ ... in theory and in practice.

Together with a critical understanding of the IID assumption, the whole class can be understood in these terms.